Transient Geodesic Alignment in Life-Like Cellular Automata on the {7,3} Heptagrid: Empirical Findings and a Structural Mechanism

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github.com/TibbinQuickcoil/heptagrid-ca

Abstract

We report an empirical investigation of two-state totalistic cellular automata on the {7,3} heptagrid — the regular tiling of the hyperbolic plane by heptagons meeting three to a vertex — implemented in the Poincaré disk model with a 7-neighbor topology. While prior work on hyperbolic cellular automata has focused predominantly on questions of computational universality, the dynamical behavior of Life-like birth-survival rule families on this tiling remains largely uncharacterized. We introduce a **geodesic alignment metric** defined as the mean angular distance between alive-cell positions and the nearest of seven primary geodesic directions through the disk center, with a random baseline of 12.857° derived analytically. Surveying eight Life-like rule families across random initial conditions on grids of depth 4 (190 cells) and depth 5 (519 cells), we find that under rule B2/S23, all twenty tested random initial conditions at both grid sizes produce minimum alignment scores below the random baseline: mean minimum 7.998° (37.8% below baseline) at depth 4, and 9.516° (26.0% below baseline) at depth 5. The effect is transient, appearing during the early growth phase and relaxing toward baseline as grid occupancy increases. We derive a structural mechanism from two properties of the {7,3} neighbor graph — concentration of geodesic adjacency along geodesic directions (7.5–11.6× ratio at depth 4) and differential shared-neighbor density between on- and off-geodesic cell pairs — both rule-independent properties of the tiling that intensify at larger scales. We also retract an initially observed period-4 oscillator under B34/S234, which does not persist at depth 5 and is a finite-grid artifact. The interactive computational explorer and all verification code are released as open-source artifacts.

1. Introduction

Conway's Game of Life [1] is among the most studied dynamical systems in mathematics. Half a century of investigation on the Euclidean square lattice has yielded a vast catalog of behaviors: still lifes, oscillators, gliders, glider guns, and computational structures sufficient to construct universal Turing machines. The square grid has become so naturalized as a substrate that the question of *which substrate* is rarely asked. Yet the choice of underlying tiling is not neutral: it determines neighborhood structure, growth rates, and the geometric constraints under which patterns evolve.

The hyperbolic plane offers a substantially different substrate. Where the Euclidean plane admits regular tilings by squares, triangles, and hexagons, the hyperbolic plane admits an infinite family of regular tilings {p, q} for which the angular constraint $(p-2)(q-2)>4$ is satisfied. Among these, the {7,3} tiling — heptagons meeting three to a vertex, the dual of the {3,7} triangular tiling — is the simplest hyperbolic regular tiling producing a 7-neighbor cell structure. As the dual of the triangular tiling, the {7,3} heptagrid has the lowest vertex valency among hyperbolic regular tilings, making it the sparsest such structure in terms of inter-cell connectivity per cell. Each cell touches exactly seven neighbors, the number of cells at graph distance n from any given cell grows exponentially rather than quadratically, and straight lines through the plane are realized as arcs of circles orthogonal to the disk boundary in the Poincaré model.

A small but significant body of work has explored cellular automata on hyperbolic tilings [2, 3]. This literature has focused almost exclusively on computational universality: whether the substrate admits cellular automata capable of simulating arbitrary Turing machines. These are top-down constructions in which transition rules are explicitly engineered, often using geodesic paths as designed communication channels.

The complementary question — what spontaneously emergent dynamical behavior arises when simple Life-like birth-survival rules are run on the {7,3} tiling — has received little attention. The closest prior exploration of which we are aware is an informal investigation by Mishin [5], conducted on the {5,4} pentagrid without quantitative analysis.

We are interested in a specific question: does the curvature of the underlying space leave detectable fingerprints on the dynamics of Life-like rules, even when those rules are themselves geometrically blind? A cellular automaton rule contains no representation of geodesics, distance, or curvature. It counts neighbors and applies a transition function. If the geometry of the substrate nevertheless biases the resulting dynamics, that bias must enter through the topology of the neighborhood graph alone.

We approach this question empirically, then mechanistically, and then critically — explicitly testing initial findings across grid sizes and retracting one that does not survive. Section 2 describes the tiling construction, CA implementation, and geodesic alignment metric. Section 3 presents the initial depth-4 survey. Section 4 derives the structural mechanism. Section 5 presents depth-5 validation and the retraction. Section 6 addresses limitations and future work.

2. Methods

2.1 Tiling Construction

We work in the Poincaré disk model, in which the hyperbolic plane is represented as the open unit disk $D = \{z \in ■ : |z|<1\}$ with metric $ds = 2|dz|/(1-|z|^2)$. Hyperbolic geodesics are diameters of the disk and arcs of circles meeting the boundary orthogonally.

The {7,3} tiling is generated by reflecting a central heptagon across its sides recursively. Two geometric constants are derived from the Schläfli symbol {7,3} using standard formulas for regular hyperbolic polygons [4]:

$$R_{■} = \operatorname{arccosh}(\cos(\pi/3) / \sin(\pi/7)) \approx 0.545 \quad \text{[circumradius: center to vertex]}$$

$$r = \operatorname{arccosh}(\cos(\pi/7) / \sin(\pi/3)) \approx 0.283 \quad \text{[inradius: center to edge midpoint]}$$

The Euclidean displacement parameter for center-to-center moves is $\rho_{cc} = \tanh(R_c)$, used to place adjacent cell centers via Möbius transforms. For rendering, vertices are placed at $\rho_v = \tanh(D/2)$ where $D = \operatorname{arccosh}(\cosh(R_c)\cdot\cosh(r)) \approx 0.621$. This parameter is used for visualization only and does not enter the CA logic or alignment metric.

Correction note. An earlier draft mislabeled R_c as 'apothem' and r as 'half-edge,' reversing the standard definitions. This was identified in peer review (see Acknowledgments). The implementation correctly uses $\rho_{cc} = \tanh(R_c)$ for cell-center placement, verified to produce a valid {7,3} neighbor graph by confirming each interior cell resolves exactly 7 neighbors.

Cell positions are enumerated by BFS beginning at the origin. Cells within radius 0.971 are retained for depth 4 (N = 190) and 0.989 for depth 5 (N = 519). All runs use fixed initial orientation $\phi = 0$, ensuring deterministic BFS order. The center cell's seven immediate neighbors (depth 1) lie exactly along the seven primary geodesic directions — consequential for metric interpretation.

2.2 Cellular Automaton

We implement two-state totalistic Life-like CA in the B/S family. The state of cell i at time t+1 is determined by its alive-neighbor count $a_i = \Sigma_{j\in N(i)} s_j$■ and birth/survival sets B, S $\subseteq$ {0,…,7}: a dead cell is born if $a_i \in$ B; a live cell survives if $a_i \in$ S; otherwise it dies. Eight rule families were surveyed: B3/S34, B2/S34, B23/S34, B3/S23, B2/S23, B34/S234, B2/S345, and B1/S12.

2.3 Geodesic Alignment Metric

For a cell at Poincaré disk coordinates (c_r, c_i), angular position $\theta = \mathrm{atan2}(c_i, c_r)$. Angular distance to the nearest of the seven primary geodesic directions:

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δ(θ) = min( t, 2π/7 − t ) where t = θ mod (2π/7)
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The geodesic alignment score is $A = (1/|F|)\ \Sigma_{c\in F}\ \delta(\theta_c) \cdot (180/\pi)$, where F excludes cells with $c_r^2+c_i^2 < 0.002$. Lower A indicates greater concentration near geodesic directions.

Random baseline. For uniformly random θ: $E[\delta] = \pi/14 \approx 12.857°$. This assumes angle-uniform rather than cell-uniform distribution; any resulting bias is symmetric across geodesic sectors and not expected to materially affect results.

Geodesic threshold. We use 7.071° ≈ 12.857° × 0.55 as an operational marker for strong geodesic bias. This is an operational choice, not derived from first principles.

Depth-1 caveat. Depth-1 cells are placed exactly at geodesic directions by the BFS construction, contributing $\delta = 0$ regardless of dynamics. Depth-stratified analysis accompanies aggregate scores throughout.

3. Initial Survey — Depth-4 Grid

3.1 Rule Survey Overview

Table 1. Rule survey results, depth-4 grid (190 cells), 20 random seeds (density 0.28), 100 generations.

Rule	Alive@100 (mean)	Mean min align	Below baseline	Behavior
B3/S34	0	—	—	extinction
B2/S34	2	8.39°	20/20	near-extinction
B23/S34	~44	11.38°	partial	diffuse oscillation
B3/S23	3	8.57°†	—	depth-0/1 cluster
B2/S23	**~38**	**7.998°**	**20/20**	**transient geodesic bias ■**
B34/S234	*36–40*	*11.44°*	*partial*	*oscillatory (retracted — §§5.2)*
B2/S345	0	—	—	extinction
B1/S12	~73	12.27°	partial	dense diffuse

† B3/S23 dominated by depth-0/1 cluster; see §2.3.

3.2 B2/S23: Transient Geodesic Alignment

Under B2/S23, all twenty random initial conditions produce minimum alignment below the 12.857° random baseline. Full distribution of minima:

4.17°, 5.71°, 7.34°, 7.35°, 7.39°, 7.48°, 7.68°, 7.75°, 7.89°, 8.01°, 8.34°, 8.35°, 8.38°, 8.40°, 8.49°, 8.68°, 9.00°, 9.28°, 9.60°, 9.61°

- Mean minimum: **7.998°** (37.8% below baseline)
- Below 7.071° threshold: 2/20
- Below baseline: 20/20

The minimum occurs at variable generations (9–71) during the intermediate-density growth phase, before relaxing toward baseline as grid occupancy increases. Figure 2 shows alignment trajectories for five representative seeds alongside the B1/S12 null result.

Depth-stratified analysis. At the global minimum (seed 7, generation 34, 12 alive cells):

Table 2. Depth-stratified alignment at minimum.

Depth	Alive cells	Alignment	Note
1	6	0.00°	geometric artifact
2	2	7.85°	sub-baseline, genuine signal
3	3	10.56°	near-baseline
4	1	2.63°	sub-baseline, small sample

Excluding depth-1 cells: 6 remaining cells at depths 2–4 show mean alignment 7.43°, below baseline. Sample sizes are small; the direction is consistent with the aggregate result.

3.3 Null Results

Five of eight rules produce no evidence of geometric organizing influence, confirming that geodesic bias is not a generic property of Life-like rules on the {7,3} tiling but a feature of specific dynamical regimes.

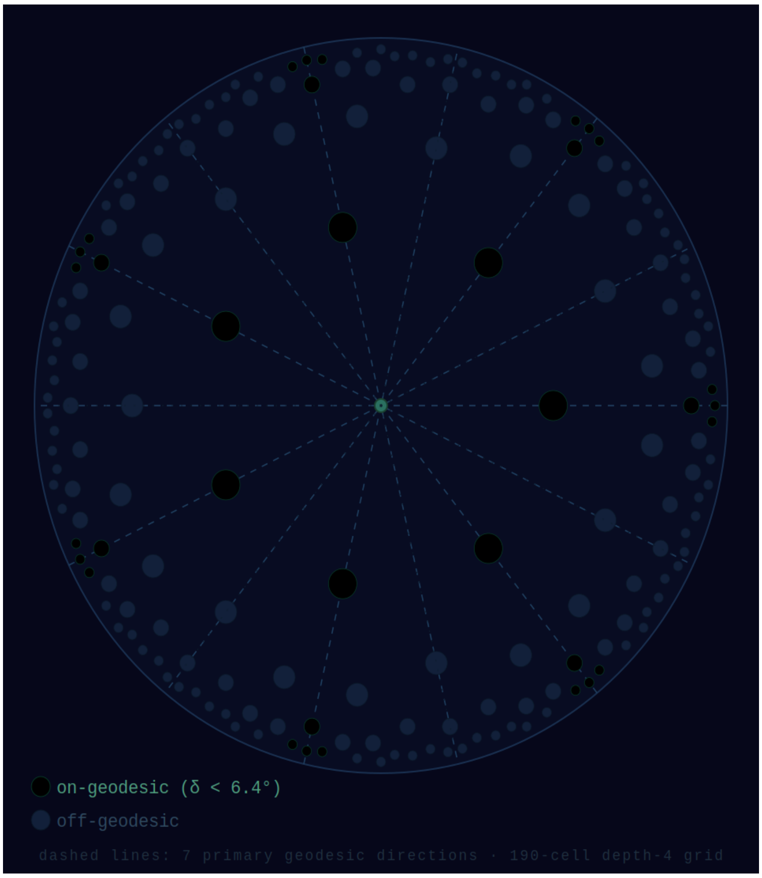

Figure 1. Geodesic classification of cells in the depth-4 {7,3} Poincaré disk. Cyan cells are on-geodesic ($\delta <$ 6.4°); gray cells are off-geodesic. Dashed lines show the seven primary geodesic directions. On-geodesic cells form elongated radial chains; off-geodesic cells form dense interstitial regions.

4. Mechanism: Structural Origin of Geodesic Bias

4.1 Approach

The transient alignment in B2/S23 admits two explanations: a statistical artifact of finite-grid effects or sampling noise, or a structural property of the {7,3} neighbor graph that systematically favors geodesic propagation for certain rule families. We test the second hypothesis directly from the neighbor graph, independent of any rule or dynamics.

4.2 Geodesic Classification

We label each cell at depth $d \geq 1$ as *on-geodesic* if $\delta(\theta_c) < \pi/28 \approx 6.43°$ — within the inner half of the angular distance to the nearest geodesic — and *off-geodesic* otherwise.

4.3 Geodesic Neighborhood Concentration

Table 3. Mean on-geodesic neighbors per cell, depth-4 grid.

Depth	Class	n	Mean on-geo neighbors	Ratio
3	on-geodesic	21	3.000	**7.5×**
3	off-geodesic	35	0.400	
4	on-geodesic	21	1.667	**10.0×**
4	off-geodesic	84	0.167	

At depth 3, on-geodesic cells have 7.5× more on-geodesic neighbors than off-geodesic cells at the same depth. At depth 4 the ratio is 10.0×. This is a purely structural property of the {7,3} tiling derived from the neighbor graph, independent of any rule or initial condition.

4.4 Shared Neighbor Density

Table 4. Mean shared neighbors among adjacent cell pairs, depth-4 grid.

Pair type	n pairs	Mean shared neighbors
Both on-geodesic	35	0.800
Both off-geodesic	161	1.043

Off-geodesic adjacent pairs share 30% more neighbors than on-geodesic pairs (1.043 vs. 0.800). The on-geodesic chain topology is sparsely cross-connected; off-geodesic regions are densely cross-connected.

4.5 The Two-Component Mechanism

Component 1 — Directed alive signal. During sparse growth, an on-geodesic cell has 7.5–10× more geodesic-adjacent neighbors than an off-geodesic cell at the same depth. If any chain cell becomes alive, its on-geodesic neighbors receive a concentrated alive-cell signal in the chain direction, biasing the probability of satisfying B2 (exactly 2 alive neighbors) along geodesics.

Component 2 — Differential birth probability. The dense cross-connectivity of off-geodesic regions pushes alive-neighbor counts above the B2 threshold, suppressing B2 birth. Crucially, this mechanism *inverts* for higher birth-count rules: for B3 or B4 rules, dense cross-connectivity facilitates rather than suppresses birth by making it easier to accumulate 3 or more simultaneous alive neighbors. This predicts a dichotomy: B2-family rules exhibit geodesic bias; B3-family rules show reduced or absent bias — consistent with our null results for all B3-based rules.

Transience. The mechanism is density-dependent. As alive-cell density increases, the structural advantage diminishes and the alignment signal relaxes toward baseline.

4.6 Status of the Argument

The structural properties in Tables 3 and 4 are computed exactly from the {7,3} neighbor graph and are rule-independent. The prediction that B2/S23 exhibits transient geodesic bias follows under a qualitative argument about sparse-field birth probabilities. A complete formal derivation is the principal direction for future theoretical work. The structural concentration ratios (7.5–10×) are too large to be explained by finite-grid noise, ruling out the sampling-artifact hypothesis.

5. Depth-5 Validation

5.1 B2/S23 at Depth 5

Table 5. B2/S23 geodesic alignment by grid depth.

Grid	N	Mean min	Reduction	Below baseline	Below 7.071°
Depth 4	190	7.998°	37.8%	20/20	2/20
Depth 5	519	9.516°	26.0%	20/20	0/20

The directional finding is confirmed: 20/20 seeds below baseline at both grid sizes. The magnitude weakens from 37.8% to 26.0%, and no depth-5 seed breaches the strong threshold.

Individual depth-5 minima: 8.80°, 8.86°, 8.99°, 9.12°, 9.12°, 9.19°, 9.26°, 9.28°, 9.34°, 9.44°, 9.47°, 9.53°, 9.58°, 9.72°, 9.82°, 9.98°, 10.02°, 10.10°, 10.34°, 10.38°.

5.2 Retraction: B34/S234 Period-4 Oscillator

At depth 4, rule B34/S234 produced a repeating alive-count cycle (39, 37, 40, 36) by approximately generation 40, persisting through generation 100 across all tested seeds. At depth 5, no such clean periodicity is observed within 120 generations. The depth-4 cycle is a finite-grid artifact specific to the 190-cell grid.

We retract this finding. The period-4 oscillator was real within the 190-cell grid; it is not a property of the rule on the infinite {7,3} tiling. This retraction is reported as an example of standard scientific practice.

5.3 Structural Mechanism at Depth 5

Table 6. Geodesic neighborhood concentration, depth-5 grid.

Depth	On-geo mean	Off-geo mean	Ratio
3	3.667	0.400	9.17×
4	5.800	0.500	11.60×
5	1.400	0.000	not computed (see text)

At depth 5, off-geodesic cells show zero on-geodesic neighbors. This is a boundary artifact: depth-5 cells have neighbors at depth 6, which fall outside the radius cutoff. The depth-5 ratio is a lower bound on the true structural concentration; division by zero is not performed. The ratios at depths 3 and 4, where both cell classes have fully resolved neighbors, show genuine intensification (7.5× to 9.17× at depth 3; 10.0× to 11.6× at depth 4).

5.4 Reconciling Stronger Mechanism with Weaker Signal

The structural concentration strengthens at depth 5 while the aggregate alignment signal weakens. This is explained by population averaging: at depth 5 the total alive-cell population is larger, and the geodesic chains form a smaller fraction of the total. The invariant result across both scales is the directional one: 20/20 seeds below baseline at both depths. This is the most defensible form of the primary finding.

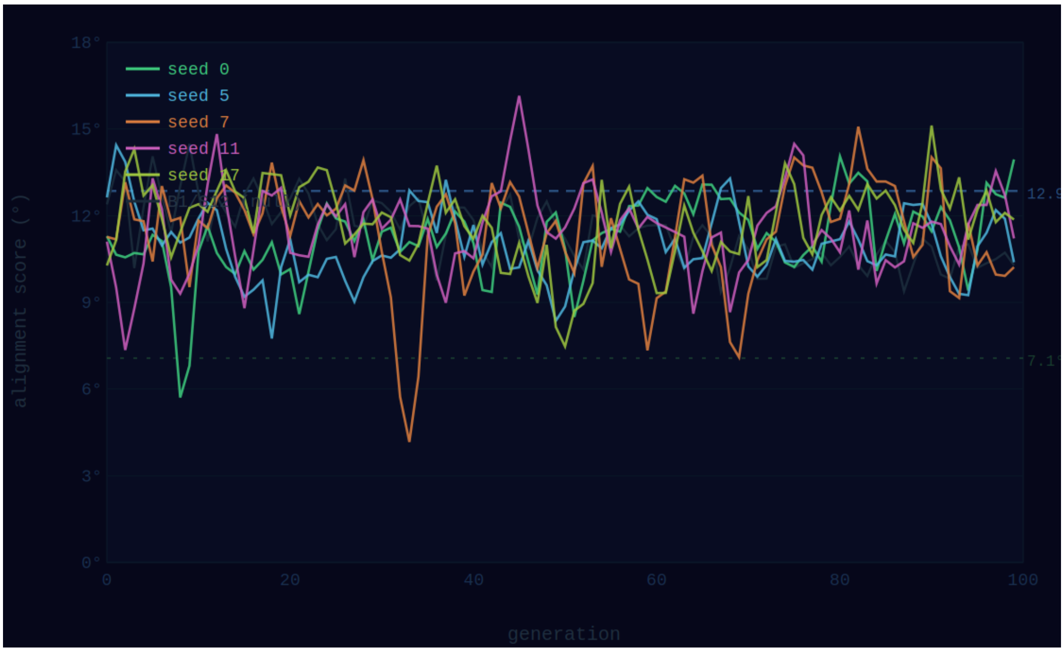

Figure 2. Geodesic alignment score over 100 generations under B2/S23 for five representative seeds (colored lines) and B1/S12 null result (dark line). Dashed line at 12.857° is the random baseline; dotted line at 7.071° is the strong bias threshold. All B2/S23 seeds descend below baseline during the growth phase before relaxing.

6. Limitations and Future Work

Formal derivation.

The mechanism in §4 is structurally grounded but not formally proven. A complete derivation would compute birth probabilities as a function of density and cell classification, yielding a quantitative prediction testable against the alignment trajectory.

Euclidean control.

Running the same rule families on a square or hexagonal grid with an analogous angular metric is the most important missing validation. This would test whether the effect is specific to hyperbolic curvature or could arise from any regular tiling with preferred directions.

Larger grids.

Depth 6 yields approximately $N \approx 1900$ cells. The population-averaging hypothesis could be tested directly via depth-stratified alignment at depth 5, and the depth-5 structural boundary artifact would be resolved.

Rule space coverage.

The full two-state totalistic rule space on a 7-neighbor grid contains 2■ × 2■ = 65,536 rules. Systematic exploration of B2-family rules is a natural extension.

Geometric constant verification.

The vertex placement parameter $\rho_v = \tanh(D/2)$ is used for rendering only. Its geometric interpretation as a standard {7,3} polygon quantity requires independent derivation.

7. Conclusion

We have presented a quantitative empirical study of Life-like cellular automata on the {7,3} heptagrid, tested at two grid scales, with a structural mechanism derived from the neighbor graph and an explicit retraction of one initial finding that did not survive scale testing.

The principal finding is that rule B2/S23 consistently produces transient sub-baseline geodesic alignment during early growth from random initial conditions: 20/20 seeds below the 12.857° random baseline at both depth 4 and depth 5, with mean minimum reductions of 37.8% and 26.0% respectively. The directional consistency across all forty tested conditions is the most defensible form of this result.

The structural mechanism — concentration of geodesic adjacency (7.5–11.6× ratio at depth 4, with genuine intensification at depth 5) and differential shared-neighbor density — is derived from properties of the {7,3} neighbor graph independent of any rule or dynamics. The divergence between a strengthening structural mechanism and a weakening aggregate alignment signal is explained as population averaging, constituting a testable prediction for future depth-stratified analysis at larger grids.

The B34/S234 period-4 oscillator does not persist at depth 5 and is retracted.

Cellular automaton rules are geometrically blind. The {7,3} tiling is not.

Acknowledgments

The interactive computational explorer and Python verification scripts were developed with AI-assisted coding tools (Anthropic Claude). Empirical observations, methodological design, analytical framing, multi-scale validation, and the decision to retract the B34/S234 finding were developed through human-AI collaboration. All results were independently verified by Python reimplementation. Peer review commentary from Gemini 2.5 contributed the suggestion to note the {7,3}/{3,7} duality and to sharpen the B3 mechanism dichotomy. Kimi identified the mislabeling of geometric constants in §2. Grok identified the numerical error in R_c and validated the analytical baseline. DeepSeek identified the table-citation inconsistency and inline citation formatting issue.

Code and Data Availability

The interactive {7,3} cellular automaton explorer, Python survey and mechanism scripts, raw data, and SVG figures are available at:

https://github.com/TibbinQuickcoil/heptagrid-ca

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